

A Radix–Carry Continuation of the Alaoglu–Erdős Program

Exact digit transducers, defect dynamics, and structural-prime constraints

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Abstract

Let

$$\vartheta = \frac{\log 3}{\log 2}, \quad \alpha = \frac{1}{\vartheta} = \frac{\log 2}{\log 3}.$$

The Alaoglu–Erdős conjecture is equivalent to the assertion that an integer value of M^ϑ , with $M \in \mathbb{N}$, forces M to be a power of 2. The attached unified report already develops the algebraic, transcendence-theoretic, determinant, finite-verification, primitive-generator, and dynamical structure of this problem. This continuation deliberately does not repeat that material. It develops a separate radix mechanism.

Write an integer in base 2 and reinterpret the same finite digit word in base 3. The resulting map Φ is an order-preserving embedding of $\mathbb{N}_0$ into the ternary Cantor integers. More strongly, it is an exact cross-prime isometry:

$$v_3(\Phi(m) - \Phi(n)) = v_2(m - n) \quad (m \neq n).$$

Its failures of additivity and multiplicativity are not errors but explicit positive carry polynomials. Reinterpreting canonical ternary digits in base 2 gives a reverse map Ψ with negative carry polynomials. These facts lead to sharp Archimedean envelopes

$$\Phi(n) \leq n^\vartheta, \quad \Psi(n) \geq n^\alpha,$$

with equality only at powers of the source base.

For a hypothetical nonintegral solution $M = 2^x$, $A = 3^x$, define

$$P_k = \Phi(M^k), \quad D_k = A^k - P_k, \quad Q_k = \Psi(A^k), \quad E_k = Q_k - M^k.$$

Then D_k, E_k are positive integers for every $k \geq 1$. This report proves exact nonlinear carry recurrences for them; an exact mantissa formula; the uniform estimate

$$\frac{\{kx\}}{3} \leq \frac{D_k}{A^k} \leq (\log 3) \{kx\};$$

a complete cross-prime valuation law for the Cantor shadow orbit P_k ; and a first-collision theorem for binary squaring which gives

$$v_3(P_{2k} - P_k^2) = 2kv_2(M) + v_2(U^k - 1), \quad M = 2^{v_2(M)}U, \quad U \text{ odd}.$$

A nonlinear spectral-radius reformulation also emerges:

$$\lim_{k \rightarrow \infty} \Phi(M^k)^{1/k} = M^\vartheta;$$

hence the original conjecture asks whether this digit-dynamical spectral radius can be an integer only on the carry-free monomial orbits $M = 2^n$.

These are rigorous paper proofs, accompanied by independent finite executable checks. They do not yet close the conjecture. The remaining obstruction is isolated as a simultaneous Archimedean/3-adic rigidity problem for the same positive defect sequence, rather than another reformulation involving only logarithms.

Status discipline. [P] means proved in this report by a complete mathematical argument. [C] means independently checked by finite computation and is used only as a sanity check. [B] refers to an input already established in the attached unified report. [T] is a proposed closing lemma or research target. No claim announced by another model or research group is used as evidence. In particular, the Alaoglu–Erdős conjecture remains open at the end of this continuation.

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1 Scope, interface to the baseline, and the new result map

The unified report is already a large consolidation. Repeating its survey of primitive generators, common odd cores, algebraic auxiliary outputs, determinant repulsion, higher differences, finite zero-free regions, and classical transcendence consequences would make this document longer while making the frontier less visible. We therefore use only the following minimal interface.

(B1) If 2^x and 3^x are positive integers and x is not an integer, then $x > 0$ and x is irrational. In fact, standard transcendence theory makes x transcendental, but irrationality is enough for the dynamical arguments below. [B]

(B2) We may put

$$M = 2^x \in \mathbb{N}, \quad A = 3^x \in \mathbb{N},$$

so that

$$A = M^\vartheta, \quad \vartheta = \log_2 3.$$

Conversely, an integer pair M, A satisfying this relation gives a solution $x = \log_2 M$. [B]

(B3) Integer translation acts by $(M, A) \mapsto (2^r M, 3^r A)$. The baseline contains much sharper primitive normalisations; below we use only this elementary compatibility. [B]

Everything after this interface is developed independently. The main new objects and their evidence levels are as follows.

Object or theorem	Content	Status
Radix embedding Φ	Binary digits re-evaluated in base 3; strict order embedding and exact 2-adic/3-adic isometry.	[P]
Carry cocycles	Exact positive correction polynomials for addition and multiplication; exact negative corrections for Ψ .	[P]
Power envelopes	$\Phi(n) \leq n^\vartheta$ and $\Psi(n) \geq n^\alpha$, with complete equality classification and explicit gaps.	[P]
Counterexample defects	Positive integer sequences D_k, E_k with exact nonlinear recurrences and carry-budget inequalities.	[P]
Mantissa observable	Exact representation of D_k/A^k by the fractional rotation $\{kx\}$ and a Cantor digit embedding; universal two-sided bounds.	[P]

Object or theorem	Content	Status
Valuation resonance	Pairwise 3-adic distances of P_k are exactly the binary distances of M^k ; outside an explicit LTE resonance set, valuations of $D_k - D_\ell$ are forced.	[P]
Square first collision	The first nonzero ternary digit of the square-carry polynomial occurs at a computable position and has coefficient 1.	[P]
Radix spectral radius	$\sup_k \Phi(M^k)^{1/k} = M^\vartheta$, giving a purely digit-dynamical equivalent form of the conjecture.	[P]
Finite executable audit	Exhaustive checks of the exact identities over large finite boxes; never used in a proof.	[C]
Closing theorem	A cross-place rigidity theorem strong enough to make the carry budget incompatible with the rotation gauge.	[T]

2 Canonical digits and radix reinterpretation

2.1 General definition

For an integer base $b \geq 2$ and $n \in \mathbb{N}_0$, write the canonical finite expansion

$$n = \sum_{i=0}^L d_i^{(b)}(n) b^i, \quad 0 \leq d_i^{(b)}(n) < b.$$

Leading zeroes are harmless, but the canonical expansion has $d_L^{(b)}(n) \neq 0$ when $n > 0$.

Definition 2.1 (Radix reinterpretation). For bases $b, c \geq 2$, define

$$\mathcal{R}_{b \rightarrow c}(n) := \sum_{i \geq 0} d_i^{(b)}(n) c^i.$$

The two maps used throughout are

$$\Phi := \mathcal{R}_{2 \rightarrow 3}, \quad \Psi := \mathcal{R}_{3 \rightarrow 2}.$$

Thus Φ replaces the place values $1, 2, 4, \dots$ of a binary word by $1, 3, 9, \dots$, while Ψ evaluates canonical ternary digits $0, 1, 2$ at the place values $1, 2, 4, \dots$

The image of Φ consists precisely of the nonnegative integers whose ternary digits belong to $\{0, 1\}$. We call them *ternary Cantor integers*. On this image, Ψ is the genuine inverse:

$$\Psi(\Phi(n)) = n.$$

On arbitrary integers, however, Ψ is not monotone and is not an inverse in either direction. For example,

$$8 = (22)_3, \quad \Psi(8) = 6, \quad 9 = (100)_3, \quad \Psi(9) = 4.$$

This warning prevents a tempting but invalid order argument later.

2.2 Strict order preservation in the expanding direction

Proposition 2.2 (Order embedding). *If $2 \leq b < c$, then $\mathcal{R}_{b \rightarrow c}$ is strictly increasing on $\mathbb{N}_0$. In particular, Φ is a strict order embedding.*

Proof. Let $m < n$, and let r be the highest digit position at which their canonical base- b expansions differ. At that position the digit of n exceeds the digit of m by at least 1. Lower positions can oppose this difference by at most

$$(b-1) \sum_{i=0}^{r-1} c^i = (b-1) \frac{c^r - 1}{c - 1} < c^r,$$

because $b < c$. Hence the contribution c^r at the highest differing position dominates every possible lower-digit loss, so $\mathcal{R}_{b \rightarrow c}(n) > \mathcal{R}_{b \rightarrow c}(m)$. $\square$

2.3 The exact cross-prime isometry

The strongest elementary feature of Φ is not its monotonicity but its preservation of the index of the first differing digit.

Theorem 2.3 (Binary–ternary valuation isometry). *For distinct $m, n \in \mathbb{N}_0$,*

$$v_3(\Phi(m) - \Phi(n)) = v_2(m - n).$$

Equivalently, the first binary digit at which m and n differ has exactly the same index as the first ternary digit at which their Cantor images differ.

Proof. Put $r = v_2(m - n)$. Congruence modulo 2^r says that the binary digits in positions $0, \dots, r-1$ agree. Failure of congruence modulo 2^{r+1} says that the digits in position r differ, necessarily by ± 1 . Therefore

$$\Phi(m) - \Phi(n) = 3^r (\pm 1 + 3z)$$

for some $z \in \mathbb{Z}$. The parenthesised factor is a 3-adic unit, proving the claim. $\square$

Corollary 2.4 (A 2-adic/3-adic Cantor embedding). *The digit formula extends continuously to*

$$\widehat{\Phi} : \mathbb{Z}_2 \longrightarrow \mathbb{Z}_3, \quad \widehat{\Phi} \left(\sum_{i \geq 0} \varepsilon_i 2^i \right) = \sum_{i \geq 0} \varepsilon_i 3^i, \quad \varepsilon_i \in \{0, 1\}.$$

It is a homeomorphism from $\mathbb{Z}_2$ onto the closed ternary Cantor subset of $\mathbb{Z}_3$, and for $u \neq v$,

$$v_3(\widehat{\Phi}(u) - \widehat{\Phi}(v)) = v_2(u - v).$$

Proof. The series converges in $\mathbb{Z}_3$. Agreement of the first r binary digits is equivalent to agreement modulo 2^r and becomes agreement of the first r ternary digits, hence agreement modulo 3^r . The argument of [Theorem 2.3](#) gives equality at the first differing digit. Compactness of $\mathbb{Z}_2$ and the explicit inverse on $\{0, 1\}$ -digit strings give the homeomorphism statement. $\square$

This isometry is the first genuinely cross-place input of the report. It turns a binary multiplicative orbit into a ternary Cantor shadow with exactly prescribed 3-adic geometry.

3 Carry cocycles: exact failure of ring homomorphism

The map Φ is neither additive nor multiplicative. Its defects are, however, completely explicit and have a fixed sign on nonnegative integers.

3.1 Addition

Let $u_i, v_i, w_i \in \{0, 1\}$ be the binary digits of $u, v, u + v$. Let $\kappa_{-1} = 0$ and let κ_i be the carry leaving position i , so

$$u_i + v_i + \kappa_{i-1} = w_i + 2\kappa_i.$$

Theorem 3.1 (Addition carry identity). *For $u, v \in \mathbb{N}_0$,*

$$\Phi(u + v) = \Phi(u) + \Phi(v) + \sum_{i \geq 0} \kappa_i 3^i.$$

Consequently Φ is superadditive, with equality exactly when the binary addition is carry-free.

Proof. Multiply the digit recurrence by 3^i and sum over i :

$$\Phi(u) + \Phi(v) + \sum_{i \geq 0} \kappa_{i-1} 3^i = \Phi(u + v) + 2 \sum_{i \geq 0} \kappa_i 3^i.$$

The shifted carry sum is $3 \sum_i \kappa_i 3^i$, giving the formula after rearrangement. Every coefficient is nonnegative, and the correction vanishes exactly when every carry vanishes. $\square$

3.2 Multiplication

Let $r_i = \sum_{p+q=i} u_p v_q$ be the raw binary convolution coefficient. On normalising the convolution to binary digits w_i , carries satisfy

$$r_i + \kappa_{i-1} = w_i + 2\kappa_i.$$

Theorem 3.2 (Multiplication carry identity). *For $u, v \in \mathbb{N}_0$,*

$$\Phi(uv) = \Phi(u)\Phi(v) + \sum_{i \geq 0} \kappa_i 3^i.$$

Thus Φ is supermultiplicative. Equality holds precisely when the raw binary convolution already has every coefficient in $\{0, 1\}$.

Proof. The same summation as in [Theorem 3.1](#) applies, with $\sum_i r_i 3^i = \Phi(u)\Phi(v)$. $\square$

Definition 3.3 (Carry cocycles). Set

$$\mathbf{a}(u, v) = \Phi(u + v) - \Phi(u) - \Phi(v), \quad \mathbf{m}(u, v) = \Phi(uv) - \Phi(u)\Phi(v).$$

On $\mathbb{N}_0^2$ these are nonnegative integers whose ternary values are the carry polynomials in [Theorems 3.1](#) and [3.2](#).

The terminology “cocycle” is useful: associativity forces identities among these corrections. For example,

$$\mathfrak{a}(u, v) + \mathfrak{a}(u + v, w) = \mathfrak{a}(v, w) + \mathfrak{a}(u, v + w).$$

Likewise, comparing two parenthesisations of a triple product yields

$$\Phi(w)\mathfrak{m}(u, v) + \mathfrak{m}(uv, w) = \Phi(u)\mathfrak{m}(v, w) + \mathfrak{m}(u, vw).$$

These identities are automatic, but they are valuable formalisation checks.

3.3 The reverse sign

For canonical ternary digits, base-3 normalisation replaces three units at position i by one unit at position $i + 1$. Evaluating at base 2 makes that operation smaller rather than larger.

Theorem 3.4 (Reverse carry identities). *For $u, v \in \mathbb{N}_0$, let λ_i be the carries in the relevant canonical base-3 addition or multiplication. Then*

$$\begin{aligned}\Psi(u + v) &= \Psi(u) + \Psi(v) - \sum_{i \geq 0} \lambda_i 2^i, \\ \Psi(uv) &= \Psi(u)\Psi(v) - \sum_{i \geq 0} \lambda_i 2^i.\end{aligned}$$

Hence Ψ is subadditive and submultiplicative.

Proof. The base-3 carry recurrence is $r_i + \lambda_{i-1} = w_i + 3\lambda_i$. Evaluate at 2 rather than 3. The shifted carry sum contributes $2 \sum_i \lambda_i 2^i$, leaving one negative copy of the carry polynomial after rearrangement. $\square$

Warning 3.5. Subadditivity and submultiplicativity do not imply monotonicity. The explicit counterexample $\Psi(8) = 6 > 4 = \Psi(9)$ remains essential. Every later use of Ψ relies on its envelope or carry formula, never on an order claim.

4 Archimedean power envelopes

The digit maps have exact growth exponents. The proof is elementary but the equality cases are the decisive feature.

4.1 The general expanding-base theorem

Theorem 4.1 (Power envelope for radix reinterpretation). *Let $2 \leq b < c$ and put*

$$\gamma = \frac{\log c}{\log b} > 1.$$

Then for every $n \in \mathbb{N}$,

$$\mathcal{R}_{b \rightarrow c}(n) \leq n^\gamma.$$

Equality holds if and only if $n = b^L$ for some $L \in \mathbb{N}_0$.

Proof. Let $L = \lfloor \log_b n \rfloor$ and write

$$Y_b = \frac{n}{b^L} = \sum_{j=0}^L d_{L-j}^{(b)}(n) b^{-j}, \quad Y_c = \frac{\mathcal{R}_{b \rightarrow c}(n)}{c^L} = \sum_{j=0}^L d_{L-j}^{(b)}(n) c^{-j}.$$

The leading terms agree, and every lower place has $c^{-j} < b^{-j}$. Therefore $Y_c \leq Y_b$. Since $Y_b \geq 1$ and $\gamma > 1$, $Y_b \leq Y_b^\gamma$. As $b^{L\gamma} = c^L$,

$$\mathcal{R}_{b \rightarrow c}(n) = c^L Y_c \leq c^L Y_b^\gamma = n^\gamma.$$

For equality, first $Y_c = Y_b$, forcing every lower digit to vanish. Then $Y_b = Y_b^\gamma$, forcing the leading digit to be 1. Thus $n = b^L$. The converse is immediate. $\square$

Theorem 4.2 (Reverse power envelope). *Under the same assumptions, put $\beta = 1/\gamma$. Then*

$$\mathcal{R}_{c \rightarrow b}(m) \geq m^\beta \quad (m \in \mathbb{N}),$$

with equality if and only if $m = c^L$.

Proof. Normalise the canonical base- c digits at their leading position:

$$Z_c = \frac{m}{c^L}, \quad Z_b = \frac{\mathcal{R}_{c \rightarrow b}(m)}{b^L}.$$

Now $b^{-j} > c^{-j}$ at every lower place, so $Z_b \geq Z_c$. Also $Z_c \geq 1 \geq Z_c^{\beta-1}$, hence $Z_c \geq Z_c^\beta$. Multiplication by $b^L = c^{L\beta}$ proves the inequality. Equality requires no lower digits and leading digit 1, exactly as before. $\square$

Specialising gives the two fundamental bounds

$$\boxed{\Phi(n) \leq n^\vartheta}, \quad \boxed{\Psi(n) \geq n^\alpha}. \quad (1)$$

4.2 Exact decomposition of the forward gap

The proof of [Theorem 4.1](#) separates two independent sources of strictness: curvature of $y \mapsto y^\vartheta$ and contraction of lower binary places when they are reweighted by powers of 3.

Proposition 4.3 (Curvature plus digit-weight gap). *Let $n \in \mathbb{N}$, $L = \lfloor \log_2 n \rfloor$, and*

$$Y_2 = \frac{n}{2^L}, \quad Y_3 = \frac{\Phi(n)}{3^L}.$$

Then

$$n^\vartheta - \Phi(n) = 3^L [(Y_2^\vartheta - Y_2) + (Y_2 - Y_3)], \quad (2)$$

$$Y_2 - Y_3 = \sum_{j=1}^L \varepsilon_{L-j} (2^{-j} - 3^{-j}), \quad (3)$$

where ε_i are the binary digits of n . In particular,

$$n^\vartheta - \Phi(n) \geq (\vartheta - 1) \left(\frac{3}{2}\right)^L (n - 2^L). \quad (4)$$

Proof. The identity is obtained by adding and subtracting $3^L Y_2$. Formula (3) is termwise. Convexity gives $Y_2^\vartheta \geq 1 + \vartheta(Y_2 - 1)$, hence $Y_2^\vartheta - Y_2 \geq (\vartheta - 1)(Y_2 - 1)$, which becomes (4) after rescaling. $\square$

The lower bound is especially strong near a power of 2: if $n = 2^L + r$ with $1 \leq r < 2^L$, then the gap is at least $(\vartheta - 1)(3/2)^L r$. Thus the integer gap in a hypothetical solution is not merely nonzero; it grows at a definite cross-radix scale.

5 Specialisation to a hypothetical counterexample

Assume from now through [Section 12](#) that

$$M = 2^x \in \mathbb{N}, \quad A = 3^x \in \mathbb{N}, \quad x \notin \mathbb{Z}. \quad (5)$$

Then $x > 0$, $M > 1$, $A > 1$, and

$$A = M^\vartheta, \quad M = A^\alpha.$$

Neither M is a power of 2 nor A a power of 3.

5.1 The four canonical sequences

Definition 5.1 (Cantor shadows and defects). For $k \geq 1$, set

$$\begin{aligned} P_k &= \Phi(M^k), & D_k &= A^k - P_k, \\ Q_k &= \Psi(A^k), & E_k &= Q_k - M^k. \end{aligned}$$

For bookkeeping at the identity element, we also set $P_0 = Q_0 = 1$ and $D_0 = E_0 = 0$.

Proposition 5.2 (Strict integer sandwich). *For every $k \geq 1$,*

$$1 \leq D_k \leq A^k - 1, \quad E_k \geq 1.$$

Equivalently,

$$\Phi(M^k) \leq A^k - 1, \quad \Psi(A^k) \geq M^k + 1.$$

Proof. By [Theorem 4.1](#), $P_k \leq (M^k)^\vartheta = A^k$. Equality would force M^k to be a power of 2, hence M a power of 2, contrary to (5). Both sides are integers, so $D_k \geq 1$. The upper bound follows from $P_k \geq 1$. The reverse assertion follows from [Theorem 4.2](#); equality would force A^k , and hence A , to be a power of 3. $\square$

5.2 Synchronous digit length

A small identity gives the radix construction its geometric coherence.

Proposition 5.3 (Equal word length). *For every $k \geq 1$, the binary word of M^k and the ternary word of A^k have the same length,*

$$L_k + 1, \quad L_k = \lfloor kx \rfloor.$$

Moreover, P_k has as its ternary word exactly the binary word of M^k . Thus $P_k < A^k$ says that the ternary word of A^k lies lexicographically above the binary word of M^k when the latter is read as a 0–1 ternary word.

Proof. Since $\log_2 M = x$ and $\log_3 A = x$,

$$\lfloor \log_2 M^k \rfloor = \lfloor kx \rfloor = \lfloor \log_3 A^k \rfloor.$$

The remaining statements are the definition of Φ and ordinary positional comparison in base 3. $\square$

This lexical dominance is exact but, by itself, is not restrictive enough: it is already implied by the universal power envelope. Its value comes from combining it with carries and valuations, not from treating it as a standalone order miracle.

5.3 Compatibility with integer translation

Let $r \geq 0$ and write $M = 2^r M_0$, $A = 3^r A_0$. Then

$$\Phi(2^r n) = 3^r \Phi(n), \quad \Psi(3^r m) = 2^r \Psi(m).$$

Consequently

$$D_k(M, A) = 3^{rk} D_k(M_0, A_0), \quad E_k(M, A) = 2^{rk} E_k(M_0, A_0).$$

Thus the radix defects respect the baseline's integer-shift reduction exactly; no information is lost by passing to a primitive translate.

6 Exact defect dynamics and the carry budget

6.1 Forward recurrence

Let $C_{k,\ell}$ denote the binary multiplication carry polynomial, evaluated at 3, for the product $M^k M^\ell = M^{k+\ell}$. By [Theorem 3.2](#),

$$P_{k+\ell} = P_k P_\ell + C_{k,\ell}, \quad C_{k,\ell} \in \mathbb{N}_0.$$

Theorem 6.1 (Forward defect recurrence). *For $k, \ell \geq 1$,*

$$\boxed{D_{k+\ell} = A^k D_\ell + A^\ell D_k - D_k D_\ell - C_{k,\ell}.} \tag{6}$$

In particular,

$$D_{2k} = 2A^k D_k - D_k^2 - C_{k,k}. \tag{7}$$

Because $D_{k+\ell} \geq 1$, every hypothetical counterexample must satisfy the carry-budget inequality

$$C_{k,\ell} + 1 \leq A^k D_\ell + A^\ell D_k - D_k D_\ell. \tag{8}$$

Proof. Substitute $P_j = A^j - D_j$ into $P_{k+\ell} = P_k P_\ell + C_{k,\ell}$ and expand. The last inequality is [Proposition 5.2](#). $\square$

The recurrence has a useful interpretation. If there were no binary carries, then $P_{k+\ell} = P_k P_\ell$, and the defect would evolve by the first three terms on the right of (6). Every actual carry consumes a positive portion of that available defect. A proof by this route must show that, at some scale, the required carry expenditure exceeds the available budget.

6.2 Reverse recurrence

Let $K_{k,\ell}$ be the base-3 multiplication carry polynomial, evaluated at 2, for $A^k A^\ell = A^{k+\ell}$. Then

$$Q_{k+\ell} = Q_k Q_\ell - K_{k,\ell}.$$

Theorem 6.2 (Reverse defect recurrence). *For $k, \ell \geq 1$,*

$$\boxed{E_{k+\ell} = M^k E_\ell + M^\ell E_k + E_k E_\ell - K_{k,\ell}.} \quad (9)$$

Hence

$$K_{k,\ell} + 1 \leq M^k E_\ell + M^\ell E_k + E_k E_\ell. \quad (10)$$

Proof. Insert $Q_j = M^j + E_j$ into the reverse carry identity and use $E_{k+\ell} \geq 1$. $\square$

The signs in (6) and (9) are genuinely different. In the forward direction the mixed defect terms are reduced by $D_k D_\ell$; in the reverse direction they are enlarged by $E_k E_\ell$. This asymmetry is one reason to retain both sequences even though the forward map has the cleaner valuation theory.

6.3 Power bounds generated by the first defect

Supermultiplicativity and submultiplicativity give immediate global bounds.

Proposition 6.3 (Propagation from $k = 1$). *For every $k \geq 1$,*

$$\begin{aligned} (A - D_1)^k &\leq P_k \leq A^k - 1, \\ 1 &\leq D_k \leq A^k - (A - D_1)^k, \\ M^k + 1 &\leq Q_k \leq (M + E_1)^k, \\ 1 &\leq E_k \leq (M + E_1)^k - M^k. \end{aligned}$$

Proof. Repeated supermultiplicativity gives $P_k = \Phi(M^k) \geq \Phi(M)^k = (A - D_1)^k$. Repeated submultiplicativity gives $Q_k \leq \Psi(A)^k = (M + E_1)^k$. The remaining bounds are definitions and [Proposition 5.2](#). $\square$

These estimates are not by themselves contradictory: the allowed intervals still have exponential width. Their main purpose is to calibrate any proposed carry lower bound.

7 Structural-prime laws

7.1 Valuation of a Cantor shadow

Let

$$a = v_2(M), \quad M = 2^a U, \quad U \text{ odd}.$$

Because the least nonzero binary digit of M^k is in position ak , the least nonzero ternary digit of P_k is in the same position.

Proposition 7.1 (Structural valuation). *For every $k \geq 1$,*

$$v_3(P_k) = ak.$$

If $U > 1$, then

$$\boxed{v_3(P_k - 3^{ak}) = ak + v_2(U^k - 1).} \quad (11)$$

The quotient after removal of this power of 3 is congruent to 1 modulo 3.

Proof. The shift identity gives $P_k = 3^{ak}\Phi(U^k)$. Since U^k is odd, its least binary digit is 1, so $\Phi(U^k)$ is a 3-adic unit. If $r = v_2(U^k - 1)$, then the first binary digit after the least digit at which U^k differs from 1 is position r , and that digit is 1. The isometry [Theorem 2.3](#), or direct digit inspection, gives $v_3(\Phi(U^k) - 1) = r$ with unit quotient 1 modulo 3. $\square$

Let $b = v_3(A)$. Since $D_k = A^k - P_k$, unequal structural slopes force the valuation of the defect without cancellation.

Corollary 7.2 (Unequal-slope defect valuation). *If $a \neq b$, then for all $k \geq 1$,*

$$v_3(D_k) = k \min(a, b).$$

If $a = b$, then $3^{ak} \mid D_k$ and

$$\frac{D_k}{3^{ak}} = \left(\frac{A}{3^a}\right)^k - \Phi(U^k).$$

Proof. When $a \neq b$, the two summands A^k and P_k have unequal 3-adic valuations bk and ak , so their difference has the smaller valuation. The equal case follows by factoring 3^{ak} . $\square$

7.2 The first collision in a binary square

The next theorem identifies not just a lower bound but the exact first ternary place occupied by the square-carry polynomial.

Theorem 7.3 (Square first-collision theorem). *Let $n = 2^a u$ with $u > 1$ odd, and set $r = v_2(u - 1)$. Then*

$$C(n) := \Phi(n^2) - \Phi(n)^2 > 0$$

and

$$\boxed{v_3(C(n)) = 2a + r, \quad \frac{C(n)}{3^{2a+r}} \equiv 1 \pmod{3}.} \quad (12)$$

Proof. The two lowest 1-bits of n occur in positions a and $a + r$. In the raw binary convolution for n^2 , the coefficient at position $2a$ is 1, and all coefficients through position $2a + r - 1$ are at most 1 with no incoming carry. At position $2a + r$, the two cross terms $(a, a + r)$ and $(a + r, a)$ contribute exactly 2. Thus the first nonzero carry leaves position $2a + r$, and that carry equals 1. The multiplication carry identity evaluates the carry string in base 3, proving (12). $\square$

Apply this to $n = M^k = 2^{ak}U^k$.

Corollary 7.4 (Exact square-carry valuation along the orbit). *If $U > 1$, then*

$$\boxed{v_3(P_{2k} - P_k^2) = 2ak + v_2(U^k - 1),} \quad (13)$$

with leading 3-adic unit congruent to 1 modulo 3. Equivalently,

$$\boxed{v_3(2A^k D_k - D_k^2 - D_{2k}) = 2ak + v_2(U^k - 1).} \quad (14)$$

Moreover,

$$D_k(2A^k - D_k) \geq D_{2k} + 3^{2ak+v_2(U^k-1)} \geq 1 + 3^{2ak+v_2(U^k-1)}. \quad (15)$$

Proof. Use [Theorem 7.3](#) and the identity $P_{2k} - P_k^2 = 2A^k D_k - D_k^2 - D_{2k}$. Positivity and the least ternary place give the real lower bound for the carry polynomial. $\square$

For $k = 2^j$, lifting the exponent gives an especially transparent law. If $U > 1$ is odd and $j \geq 1$, then

$$v_2(U^{2^j} - 1) = v_2(U - 1) + v_2(U + 1) + j - 1.$$

Therefore

$$v_3(P_{2^{j+1}} - P_{2^j}^2) = 2a2^j + j + \sigma_U, \quad \sigma_U = v_2(U - 1) + v_2(U + 1) - 1. \quad (16)$$

This exact affine–exponential valuation profile is a compact target for Lean formalisation and for any future 3-adic interpolation argument.

7.3 Pairwise valuation geometry and the resonance set

The square theorem is one slice of a complete pairwise law.

Theorem 7.5 (Pairwise Cantor-shadow distances). *For distinct $k, \ell \geq 0$,*

$$\boxed{v_3(P_k - P_\ell) = v_2(M^k - M^\ell).} \quad (17)$$

Proof. This is [Theorem 2.3](#) applied to M^k and M^ℓ . $\square$

For $k > \ell$, define

$$R_{k,\ell} = v_2(M^k - M^\ell), \quad S_{k,\ell} = v_3(A^k - A^\ell).$$

Since

$$P_k - P_\ell = (A^k - A^\ell) - (D_k - D_\ell),$$

the ultrametric inequality determines the defect difference whenever the two known valuations are unequal.

Theorem 7.6 (Valuation resonance dichotomy). *If $R_{k,\ell} \neq S_{k,\ell}$, then*

$$\boxed{v_3(D_k - D_\ell) = \min(R_{k,\ell}, S_{k,\ell}).} \quad (18)$$

If $S_{k,\ell} < R_{k,\ell}$, then additionally

$$D_k - D_\ell \equiv A^k - A^\ell \pmod{3^{R_{k,\ell}}}.$$

All exceptional cancellation is therefore confined to the explicit resonance locus

$$R_{k,\ell} = S_{k,\ell}.$$

Proof. Set $X = A^k - A^\ell$ and $Y = D_k - D_\ell$. We know $v_3(X - Y) = R_{k,\ell}$ and $v_3(X) = S_{k,\ell}$. If these valuations are unequal, the non-Archimedean triangle law forces $v_3(Y)$ to be their minimum. When $S < R$, the relation $Y = X - (X - Y)$ also gives the displayed congruence. $\square$

The resonance locus can be written explicitly by LTE. Let $d = k - \ell$. If $a = v_2(M) > 0$, then

$$R_{k,\ell} = a\ell,$$

because $M^d - 1$ is odd. If $a = 0$, so M is odd, then

$$R_{k,\ell} = \begin{cases} v_2(M - 1), & d \text{ odd}, \\ v_2(M - 1) + v_2(M + 1) + v_2(d) - 1, & d \text{ even}. \end{cases}$$

Similarly, if $b = v_3(A) > 0$, then $S_{k,\ell} = b\ell$. If $b = 0$, then

$$S_{k,\ell} = \begin{cases} v_3(A - 1) + v_3(d), & A \equiv 1 \pmod{3}, \\ 0, & A \equiv -1 \pmod{3} \text{ and } d \text{ odd}, \\ v_3(A^2 - 1) + v_3(d), & A \equiv -1 \pmod{3} \text{ and } d \text{ even}. \end{cases}$$

Thus resonance is not an amorphous exceptional set: it lies on finitely many linear relations among $v_2(d)$, $v_3(d)$, and ℓ .

8 The mantissa observable and a sharp rotation gauge

This section links the real size of D_k to the irrational rotation $k \mapsto \{kx\}$. Unlike a generic asymptotic estimate, the initial formula is exact.

8.1 A finite Cantor embedding of a dyadic mantissa

Put

$$L_k = \lfloor kx \rfloor, \quad t_k = \{kx\} \in (0, 1).$$

Then

$$M^k = 2^{L_k + t_k} = 2^{L_k}(1 + u_k), \quad u_k = 2^{t_k} - 1.$$

Because M^k is an integer, u_k is a dyadic rational with a terminating binary expansion of length at most L_k :

$$u_k = \sum_{j=1}^{L_k} \varepsilon_{k,j} 2^{-j}.$$

Define its finite Cantor embedding

$$\mathcal{C}(u_k) = \sum_{j=1}^{L_k} \varepsilon_{k,j} 3^{-j}.$$

The normalised Cantor shadow is then

$$\frac{P_k}{3^{L_k}} = 1 + \mathcal{C}(u_k),$$

whereas

$$\frac{A^k}{3^{L_k}} = 3^{t_k}.$$

Theorem 8.1 (Exact mantissa formula). *For every $k \geq 1$,*

$$\boxed{\frac{P_k}{A^k} = 3^{-t_k} (1 + \mathcal{C}(2^{t_k} - 1))}, \quad (19)$$

and hence

$$\boxed{\frac{D_k}{A^k} = 1 - 3^{-t_k} (1 + \mathcal{C}(2^{t_k} - 1))}. \quad (20)$$

Proof. The binary digits below the leading bit of M^k are exactly the terminating binary digits of u_k . Reinterpreting them in base 3 gives $P_k = 3^{L_k} (1 + \mathcal{C}(u_k))$. Divide by $A^k = 3^{L_k + t_k}$. $\square$

8.2 Universal two-sided bounds

Termwise, $3^{-j} \leq 2^{-j}$, so

$$0 \leq \mathcal{C}(u_k) \leq u_k = 2^{t_k} - 1.$$

This loses the fine digit structure but produces sharp, uniform control.

Theorem 8.2 (Rotation gauge for the forward defect). *For every $k \geq 1$,*

$$3^{-t_k} \leq \frac{P_k}{A^k} \leq \left(\frac{2}{3}\right)^{t_k}, \quad (21)$$

so

$$1 - \left(\frac{2}{3}\right)^{t_k} \leq \frac{D_k}{A^k} \leq 1 - 3^{-t_k}. \quad (22)$$

In particular,

$$\boxed{\frac{t_k}{3} \leq \frac{D_k}{A^k} \leq (\log 3) t_k}. \quad (23)$$

Proof. The bounds in (21) follow from $1 \leq 1 + \mathcal{C}(u_k) \leq 1 + u_k = 2^{t_k}$. Subtract from 1 to obtain (22). The function $f(t) = 1 - (2/3)^t$ is concave on $[0, 1]$, with $f(0) = 0$ and $f(1) = 1/3$, so $f(t) \geq t/3$. Finally $1 - e^{-y} \leq y$ with $y = t \log 3$ gives the upper linear bound. $\square$

Equation (23) is one of the central new reductions. It says that the same integer D_k is simultaneously

- an exact positive carry-budget variable in (6);
- a 3-adically constrained difference in [Theorem 7.6](#) and [Corollary 7.4](#); and
- an Archimedean gauge of the rotation distance $t_k = \{kx\}$, within absolute constants independent of the hypothetical pair.

8.3 Density and oscillation

Because x is irrational, the sequence $t_k = \{kx\}$ is dense in $[0, 1]$. The gauge immediately gives a nontrivial oscillation law.

Corollary 8.3 (Macroscopic and thin relative defects). *For a hypothetical counterexample,*

$$\liminf_{k \rightarrow \infty} \frac{D_k}{A^k} = 0,$$

and

$$\frac{1}{3} \leq \limsup_{k \rightarrow \infty} \frac{D_k}{A^k} \leq \frac{2}{3}.$$

Proof. Choose a subsequence with $t_k \rightarrow 0$ and use the upper bound in (23). Choose another with $t_k \rightarrow 1$ and use the lower bound in (22). The universal upper bound follows from $1 - 3^{-t} < 2/3$ for $t < 1$. $\square$

Thus D_k is never a perturbative error sequence in absolute size. It has subsequences that are small relative to A^k , but it also repeatedly occupies a fixed positive fraction of A^k .

8.4 The reverse mantissa

For completeness, write

$$A^k = 3^{L_k}(1 + v_k), \quad v_k = 3^{t_k} - 1,$$

and let

$$\mathcal{G}(v_k) = \sum_{j=1}^{L_k} \eta_{k,j} 2^{-j}$$

be the canonical ternary fractional digits of v_k , evaluated at binary fractional place values. Then

$$\frac{Q_k}{M^k} = 2^{-t_k}(1 + \mathcal{G}(3^{t_k} - 1)).$$

Since $2^{-j} \geq 3^{-j}$ termwise, $\mathcal{G}(v_k) \geq v_k$, and therefore

$$\frac{E_k}{M^k} \geq \left(\frac{3}{2}\right)^{t_k} - 1 \geq (\log(3/2))t_k. \quad (24)$$

An elementary first-nonzero-digit estimate also gives

$$\mathcal{G}(v) \leq 4v^\alpha \quad (0 < v < 2),$$

so, using $3^t - 1 \leq 2t$ on $[0, 1]$,

$$\frac{E_k}{M^k} \leq 4(2t_k)^\alpha. \quad (25)$$

The reverse defect is therefore controlled by a Hölder, rather than a linear, window near $t = 0$. This loss reflects the digit 2 and the nonmonotonicity of Ψ .

9 Diophantine consequences of the rotation gauge

9.1 An elementary absolute lower bound

Let $L = L_k$. Since M^k is an integer strictly larger than 2^L ,

$$2^{t_k} - 1 = \frac{M^k - 2^L}{2^L} \geq 2^{-L}.$$

Convexity of 2^t on $[0, 1]$ gives $2^t \leq 1 + t$, hence $t_k \geq 2^{-L}$. Combining with (23) yields

$$D_k \geq \frac{A^k}{3 \cdot 2^{L_k}} \geq \frac{1}{3} \left(\frac{3}{2}\right)^{L_k}.$$

The sharper digit-sensitive bound (4) is

$$D_k \geq (\vartheta - 1) \left(\frac{3}{2}\right)^{L_k} (M^k - 2^{L_k}).$$

Both are unconditional consequences of the radix envelope once the output A^k is integral.

9.2 Baker–Matveev amplification

The elementary bound allows t_k to be exponentially small in k . For a fixed integer M not equal to a power of 2, linear forms in logarithms rule out such extreme approximation.

Theorem 9.1 (Effective polynomial lower bound). *Fix an integer $M > 1$ that is not a power of 2, and put $x = \log_2 M$. There exist effectively computable constants $c(M) > 0$ and $C(M) > 0$ such that for every $k \geq 1$,*

$$\{kx\} \geq c(M)k^{-C(M)}.$$

Consequently, if $A = M^\vartheta$ is an integer, then

$$D_k \geq \frac{c(M)}{3} A^k k^{-C(M)}. \quad (26)$$

Proof. Let $L_k = \lfloor kx \rfloor$. The nonzero linear form

$$\Lambda_k = k \log M - L_k \log 2 = (\log 2) \{kx\}$$

has fixed algebraic logarithms and integer coefficients of size $O(k)$. A standard explicit lower bound of Baker–Wüstholz or Matveev gives $|\Lambda_k| \geq c'(M)k^{-C(M)}$ after adjusting constants for finitely many small k . Apply (23). $\square$

The point is qualitative but important: on every scale, the positive integer D_k is exponentially large up to a fixed polynomial factor. Therefore a closing argument cannot treat D_k as a small absolute error. It must use its special digit or 3-adic structure.

9.3 Infinitely many thin relative defects

Continued-fraction convergents to an irrational number alternate around it. There are therefore infinitely many k with

$$0 < \{kx\} < \frac{1}{k}.$$

For those k , the upper gauge gives

$$1 \leq D_k < \frac{\log 3}{k} A^k. \quad (27)$$

Thus the same sequence has Baker-controlled lower size at every index and Dirichlet-thin relative size on an infinite subsequence. Any successful cross-place argument should be tested first on this subsequence.

10 A nonlinear spectral-radius reformulation

The carry inequalities naturally suggest a spectral point of view.

Definition 10.1 (Radix spectral radii). For $n, m \geq 1$, define

$$r_\Phi(n) = \lim_{k \rightarrow \infty} \Phi(n^k)^{1/k}, \quad r_\Psi(m) = \lim_{k \rightarrow \infty} \Psi(m^k)^{1/k},$$

provided the limits exist.

Theorem 10.2 (Exact radix spectral radii). *For every $n, m \in \mathbb{N}$,*

$$r_\Phi(n) = n^\vartheta = \sup_{k \geq 1} \Phi(n^k)^{1/k}, \quad (28)$$

$$r_\Psi(m) = m^\alpha = \inf_{k \geq 1} \Psi(m^k)^{1/k}. \quad (29)$$

Proof. The upper envelope gives $\Phi(n^k) \leq n^{k\vartheta}$. If $L = \lfloor \log_2 n^k \rfloor$, then $\Phi(n^k) \geq 3^L > n^{k\vartheta}/3$. Taking k th roots squeezes the limit to n^ϑ . Supermultiplicativity of Φ makes $\log \Phi(n^k)$ superadditive in k , so Fekete's lemma identifies the limit with the supremum of the normalised terms.

For Ψ , the reverse envelope gives the lower bound $m^{k\alpha}$. If the ternary length is $L + 1$, then

$$\Psi(m^k) \leq 2 \sum_{i=0}^L 2^i < 2^{L+2} < 4m^{k\alpha}.$$

The k th roots again squeeze the limit. Submultiplicativity and Fekete's lemma for subadditive logarithms give the infimum formula. $\square$

Corollary 10.3 (Spectral-radius form of Alaoglu–Erdős). *The Alaoglu–Erdős conjecture is equivalent to*

$$r_\Phi(M) \in \mathbb{N} \implies M = 2^j \text{ for some } j \in \mathbb{N}_0.$$

Equivalently: the multiplicative orbit of an integer has an integer binary-to-ternary radix spectral radius only when every power belongs to the carry-free one-bit orbit.

Proof. By [Theorem 10.2](#), $r_\Phi(M) = M^\vartheta$. If this is the integer A , then with $x = \log_2 M$ we have $2^x = M$ and $3^x = M^\vartheta = A$. The original conjecture says that x is integral, which is equivalent to M being a power of 2. $\square$

This reformulation is not presented as a proof by renaming. Its value is that $P_k = \Phi(M^k)$ is generated by a fixed multiplication-by- M digit transducer, while $r_\Phi(M)$ is its orbit growth rate under the Archimedean weight 3^i . It opens a route through transfer operators, pressure, or infinite-state Perron–Frobenius theory that is absent from a logarithm-only formulation.

11 The adelic shadow orbit

The construction can be summarised as a single orbit viewed at two places.

11.1 Two compatible geometries

The binary orbit

$$1, M, M^2, M^3, \dots \subset \mathbb{Z}_2$$

has the Cantor shadow

$$1, P_1, P_2, P_3, \dots \subset \mathbb{Z}_3, \quad P_k = \widehat{\Phi}(M^k).$$

By [Theorem 7.5](#), all pairwise distances are transported exactly:

$$|P_k - P_\ell|_3 = 3^{-v_2(M^k - M^\ell)}.$$

At the Archimedean place the shadow sits strictly below the geometric orbit A^k :

$$0 < P_k < A^k, \quad A^k - P_k = D_k.$$

The same D_k is then governed by the rotation gauge (23).

This produces an exact three-way dictionary:

$$\begin{array}{ccc} M^k \in \mathbb{Z}_2 & \xrightarrow{\widehat{\Phi}} & P_k \in \mathcal{C}_3 \subset \mathbb{Z}_3 \\ & & \downarrow +D_k \\ & & A^k \in \mathbb{Z} \subset \mathbb{R} \cap \mathbb{Z}_3. \end{array}$$

The horizontal arrow is an ultrametric isometry; the vertical arrow is a positive Archimedean displacement with exact nonlinear recurrences.

11.2 Why this is more than the compact-orbit reformulation

The baseline already records compact and S -integral orbit formulations. The radix shadow adds two pieces of data not supplied by compactness alone:

1. the point P_k is not merely close to an orbit; it belongs to a rigid Cantor subset and carries the complete binary digit word of M^k ;
2. the valuation of every difference $P_k - P_\ell$ is known exactly, not just bounded.

This exactness is what makes [Theorems 7.3](#) and [7.6](#) possible. The unresolved task is to make the Archimedean order $P_k < A^k$ interact decisively with the ultrametric isometry.

12 Audited dead ends and the precise missing step

The radix mechanism is strong enough to generate seductive false proofs. It is useful to record exactly where they fail.

12.1 The envelope alone is universal

The inequality $\Phi(M^k) \leq A^k$ may look like a new restriction on a counterexample. It is not: $\Phi(n) \leq n^\vartheta$ holds for every integer n . The genuinely arithmetic input is that A^k is an integer, so D_k is an integer, and that all D_k belong to one coupled carry recurrence. Any argument using only the pointwise envelope is a reformulation, not a closure.

12.2 The reverse map is not monotone

From $P_k < A^k$ one cannot apply Ψ and retain the inequality. The example $8 < 9$ but $\Psi(8) > \Psi(9)$ is the smallest visible obstruction. The valid reverse inequality $\Psi(A^k) > M^k$ comes independently from the reverse power envelope.

12.3 Positive carries can absorb the entire discrepancy

Equation (6) does not make $D_{k+\ell}$ grow monotonically. A large carry polynomial subtracts from the nominal defect budget. Conversely, the fact that the carry is large is not itself a contradiction: its real size can be exponential while its first nonzero ternary place is only logarithmic in k .

12.4 Large 3-adic divisibility does not imply real smallness

The exact valuation

$$v_3(P_{2k} - P_k^2) = 2ak + v_2(U^k - 1)$$

can grow without forcing the positive integer $P_{2k} - P_k^2$ to vanish. A product-formula argument would need an independent upper bound below the corresponding power of 3. The current Archimedean bounds are far too large for that.

12.5 The two defects do not automatically descend

Both $D_k < A^k$ and, from a crude digit bound, $E_k < 2M^k$ for large scales. This suggests a descent from (M^k, A^k) to (E_k, D_k) . But no identity $D_k = E_k^\vartheta$ is available, and the carry corrections prevent the two radix maps from forming an order adjunction on all integers. A descent claim without a preserved exponential relation is invalid.

12.6 A precise closing lemma

The new material reduces the desired contradiction to a cross-place rigidity statement of the following kind.

Research problem 12.1 (Carry-rotation rigidity). *Classify integer pairs $M, A > 1$ for which there exist positive integer sequences D_k and nonnegative carry polynomials $C_{k,\ell}$ satisfying simultaneously*

- (i) $A = M^\vartheta$;
- (ii) the exact recurrence (6);
- (iii) the rotation gauge (23);
- (iv) the cross-prime isometry constraints (17)–(18); and
- (v) the first-collision law (14) with leading unit 1.

Prove that the only possibilities have $M = 2^j$ and $A = 3^j$.

Condition (i) is still the original transcendental relation, so **Research problem 12.1** is not advertised as a logically weaker theorem. The point is methodological: conditions (ii)–(v) expose rigid, computable consequences that a proof may attack by p -adic interpolation, transfer operators, or carry combinatorics. The baseline formulations do not package these consequences in one sequence.

13 Promising next attacks

13.1 A transfer operator for multiplication by M

Multiplication by a fixed M is a finite-carry transducer when binary digits are read from least significant to most significant. Iterating it generates the words of M^k . The functional

$$w \mapsto \sum_i w_i 3^i$$

assigns an Archimedean weight to the output word. The exact spectral radius [Theorem 10.2](#) suggests constructing an infinite-dimensional positive transfer operator whose orbit vector is P_k . The desired rigidity would follow from a theorem of the schematic form:

Conjectural target 13.1 (Integer pressure rigidity). *If the orbit pressure of the multiplication-by- M binary carry transducer, under ternary place weights, is the logarithm of an integer, then the transducer orbit from 1 is eventually carry-free. Consequently M is a power of 2.*

A finite matrix representation is not presently justified; claiming one would be false in general because digit sums of powers need not satisfy a fixed linear recurrence. The plausible object is a transfer operator on a space of carry-state distributions or 2-adic cylinders.

13.2 Exploit the sparse resonance locus

Outside $R_{k,\ell} = S_{k,\ell}$, the valuation of $D_k - D_\ell$ is exact. The LTE formula shows that resonance lies on explicit diagonals in $(v_2(k - \ell), v_3(k - \ell), \ell)$. A possible route is:

1. choose many indices with prescribed 2- and 3-adic valuations but with fractional parts $\{kx\}$ in a short real interval;
2. use [\(23\)](#) to confine the corresponding D_k to short Archimedean intervals;
3. use [Theorem 7.6](#) to force their pairwise differences into incompatible 3-adic residue classes.

The difficulty is quantitative. Ordinary equidistribution gives real intervals of polynomial resolution, whereas forcing a positive integer to vanish from 3-adic divisibility needs exponential resolution. A successful version must gain cancellation by combining many differences, not by a single pair.

13.3 Carry-polynomial S -unit estimates

The square carry

$$C_k = P_{2k} - P_k^2$$

has a ternary expansion whose first nonzero digit is known exactly. Its remaining digits are generated by a constrained convolution of the binary support of M^k . One can ask whether infinitely many unusually sparse carry polynomials are possible when M has an odd factor. Results on digit sums of powers and S -unit equations suggest the following intermediate target.

Research problem 13.2 (Sparse-carry exclusion). *Prove an effective lower bound for the number or weighted span of nonzero ternary digits of C_k for M not a power of 2, strong enough to conflict with the carry budget [\(8\)](#) on a Dirichlet-thin subsequence.*

A mere statement that the digit sum tends to infinity is too weak; the budget is exponential. The needed estimate must retain the position and multiplicity of carries.

13.4 A two-place determinant

The baseline’s determinant methods are Archimedean. The isometry [Theorem 2.3](#) makes it natural to build determinants from several $P_k = A^k - D_k$ such that

- row or column differences acquire prescribed high powers of 3 from the binary orbit; while
- the Archimedean size is reduced by choosing k with clustered fractional parts $\{kx\}$.

The target is a nonzero integer determinant whose real absolute value is less than 1, with its nonvanishing certified by the first differing Cantor digit. This is more specific than a generic Vandermonde repetition: the 3-adic valuation matrix is known exactly from $M^k - M^\ell$.

13.5 Certified computation as theorem discovery

A useful search should track more than near-integrality of M^θ . For each candidate integer M and its only possible nearby A , compute

$$D_1, D_2, D_4, \dots,$$

the exact carry polynomials, their first-collision positions, and the resonance matrix $R_{k,\ell} - S_{k,\ell}$. The goal is to discover a finite inequality or congruence pattern that can later be proved symbolically. This is complementary to the baseline’s directed-rounding scan and should not be reported as a larger zero-free theorem until certificates are kernel-replayed.

14 Lean formalisation plan

The results above are designed to separate cleanly into small modules. The following dependency order avoids importing the large existing corpus until the final candidate specialisation.

14.1 Module 1: canonical digit evaluation

Suggested definitions:

```
-- schematic Lean, not claimed to compile verbatim

def evalDigits (base : Nat) (d : List Nat) : Nat :=
  d.foldr (fun x acc => x + base * acc) 0

def reinterpret (source target n : Nat) : Nat :=
  evalDigits target (Nat.digits source n)

def phi : Nat -> Nat := reinterpret 2 3

def psi : Nat -> Nat := reinterpret 3 2
```

Core theorems are canonical-digit reconstruction, shift compatibility, and strict monotonicity of `reinterpret b c` under $b < c$.

14.2 Module 2: valuation isometry

Use integers for subtraction:

```
theorem padicVal_phi_sub
  {m n : Nat} (h : m != n) :
  padicValInt 3 ((phi m : Int) - phi n)
  = padicValInt 2 ((m : Int) - n)
```

The proof should avoid general p -adic analysis: identify the least differing digit and factor the difference as 3^r times a unit.

14.3 Module 3: carry normalisation

Define a generic normaliser for a finite coefficient list in base b , returning both output digits and carries. Prove a polynomial-evaluation identity once:

$$\text{eval}_c(\text{normalise}_b r) = \text{eval}_c(r) + (c - b) \text{eval}_c(\text{carry}).$$

Addition and multiplication are then corollaries. This general statement is less brittle than separate bit-level proofs.

14.4 Module 4: real envelopes

The only analytic inputs are log, real powers, and convexity of $y \mapsto y^\gamma$ on $[1, \infty)$. A useful intermediate lemma is the normalised digit comparison

$$\frac{\mathcal{R}_{b \rightarrow c}(n)}{c^L} \leq \frac{n}{b^L}.$$

Equality classification should be formalised at the same time; downstream strict positivity of D_k depends on it.

14.5 Module 5: candidate defects

Import the existing candidate predicate only here. Define P_k, D_k, Q_k, E_k and prove the recurrences by ring normalisation. The carry-budget inequalities then require only positivity of the strict defects.

14.6 Module 6: first collision and LTE

Formalise the two lowest set-bit positions of $2^a u$. The key local statement is that all square-convolution coefficients before $2a + r$ normalise without a carry and the coefficient at $2a + r$ is exactly 2. Combine this with Mathlib's LTE lemmas to obtain (16).

14.7 Module 7: mantissa gauge

Avoid formalising a global discontinuous Cantor function initially. For each k , use the finite digit list of M^k and define its normalised binary and ternary evaluations. The inequalities

$$1 \leq Y_3 \leq Y_2 = 2^{\{kx\}}$$

are sufficient for (23). The exact Cantor formula can be added as an interface theorem after the finite proof is stable.

14.8 Proposed theorem inventory

A compact first milestone is:

Suggested Lean name	Mathematical statement
<code>phi_strictMono</code>	Proposition 2.2 for bases 2, 3.
<code>padicVal_phi_sub_eq</code>	Theorem 2.3 .
<code>phi_add_eq_add_carry</code>	Theorem 3.1 .
<code>phi_mul_eq_mul_carry</code>	Theorem 3.2 .
<code>psi_add_eq_sub_carry</code>	addition part of Theorem 3.4 .
<code>psi_mul_eq_sub_carry</code>	multiplication part of Theorem 3.4 .
<code>phi_le_rpow</code>	$\Phi(n) \leq n^\vartheta$.
<code>phi_eq_rpow_iff</code>	equality iff n is a power of 2.
<code>psi_ge_rpow</code>	$\Psi(n) \geq n^\alpha$.
<code>candidate_defect_pos</code>	Proposition 5.2 .
<code>defect_mul_recurrence</code>	(6).
<code>squareCarry_padicVal</code>	Theorem 7.3 .
<code>candidate_defect_rotation_bounds</code>	(23).
<code>radixSpectralRadius</code>	Theorem 10.2 .
<code>alaogluErdos_iff_integer_radixRadix</code>	Corollary 10.3 .

None of these identifiers is claimed to exist in the current repository. The list is a formalisation specification, not a status report.

15 Independent executable audit

A short Python program was used only to detect indexing and sign mistakes. The exact proofs above do not depend on it. The audit performed the following checks:

- [Theorem 2.3](#) for every pair $0 \leq n < m < 2048$;
- both addition and multiplication carry identities, in both radix directions, for every operand pair $0 \leq u, v < 384$;
- [Theorem 7.3](#) for every $1 < n < 250000$ that is not a power of 2;
- the two real envelopes for every $1 \leq n \leq 200000$, using 100-decimal-digit arithmetic as a numerical sanity check.

All checks passed. [C] The central routines are reproduced below so that the sign convention is auditable.

```
def digits(n, base):
    if n == 0:
        return [0]
    out = []
    while n:
        n, r = divmod(n, base)
        out.append(r)          # least significant first
    return out
```

```

def eval_digits(ds, base):
    acc = 0
    for d in reversed(ds):
        acc = acc * base + d
    return acc

def reinterpret(n, source, target):
    return eval_digits(digits(n, source), target)

def mul_carry_poly(u, v, source, target):
    du, dv = digits(u, source), digits(v, source)
    raw = [0] * (len(du) + len(dv) + 4)
    for i, a in enumerate(du):
        for j, b in enumerate(dv):
            raw[i + j] += a * b
    carry = 0
    poly = 0
    for i in range(len(raw)):
        total = raw[i] + carry
        carry = total // source
        poly += carry * target**i
    return poly

phi = lambda n: reinterpret(n, 2, 3)
psi = lambda n: reinterpret(n, 3, 2)

# Forward sign: positive carry polynomial.
assert phi(u*v) == phi(u)*phi(v) + mul_carry_poly(u,v,2,3)
# Reverse sign: negative carry polynomial.
assert psi(u*v) == psi(u)*psi(v) - mul_carry_poly(u,v,3,2)

```

For a kernel-verified development, the finite audit should be replaced by the module plan of the previous section, not imported through `native_decide` as a substitute for the general proofs.

16 Worked symbolic examples

16.1 A single binary carry

Take $u = v = 1$. Binary addition gives $1 + 1 = 10_2$. Hence

$$\Phi(2) = 3, \quad \Phi(1) + \Phi(1) = 2,$$

and the carry polynomial is 1. The difference $3 - 2$ is exactly the gain from replacing two units of 3^0 by one unit of 3^1 .

16.2 A carry-free and a carrying square

For $n = 5 = (101)_2$,

$$\Phi(5) = 10, \quad 25 = (11001)_2, \quad \Phi(25) = 109.$$

Thus

$$C(5) = 109 - 10^2 = 9 = 3^2.$$

Here $v_2(5 - 1) = 2$, agreeing with [Theorem 7.3](#).

For $n = 2^a$, every power has one binary digit, so

$$\Phi(n^2) = \Phi(n)^2$$

and the carry polynomial vanishes. The equality case of the power envelope identifies precisely these monomial orbits.

16.3 Why the reverse map can fall

The canonical ternary increment

$$(22)_3 + 1 = (100)_3$$

contains a two-position carry cascade. Evaluating before normalisation at base 2 gives $6 + 1 = 7$; evaluating after normalisation gives 4. The reverse carry polynomial is 3, exactly accounting for the loss. This is the local mechanism behind $\Psi(8) = 6 > 4 = \Psi(9)$.

17 Conclusions

This continuation does not produce a completed proof of the Alaoglu–Erdős conjecture. It does produce a coherent new rigidity package that is independent of the baseline’s main routes:

1. The binary-to-ternary digit map is simultaneously an Archimedean order embedding and an exact 2-adic/3-adic isometry.
2. Addition and multiplication have explicit positive carry cocycles; the reverse digit evaluation has explicit negative cocycles.
3. The smooth curve $y = x^\vartheta$ is an exact upper envelope of the Cantor graph $y = \Phi(x)$, touching it only at powers of 2.
4. A hypothetical counterexample generates positive integer defects D_k, E_k with coupled nonlinear recurrences.
5. The relative forward defect is comparable, with absolute constants, to $\{kx\}$, while its pairwise 3-adic geometry is inherited exactly from the binary power orbit.
6. The first square carry has a computable location and unit coefficient, producing exact valuation and real carry-budget constraints.
7. The conjecture is equivalent to integer rigidity of a nonlinear radix spectral radius.

The most promising next move is not another isolated inequality. It is a multi-index construction that combines the exact 3-adic distance matrix of the Cantor shadows with clustered real rotation phases, so that many defect differences cancel Archimedeanly while retaining certified nonzero 3-adic leading terms. That is a concrete determinant/interpolation problem and a natural target for both Lean formalisation and experimental theorem discovery.

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